

Session 1

Programming Before Python

STUDENT HANDOUT

A simple foundation before your first Python code.

Goal: Understand what programming is, how computers follow instructions, why programming languages exist, and where Python fits.

Remember:
Python is a tool for programming. Programming is problem-solving.

WHAT YOU WILL LEARN

01	What a computer does
02	Hardware vs software
03	Programming vs coding
04	Algorithms and instructions
05	Why programming languages exist
06	Where Python came from
07	How Python code gets executed
08	The building blocks of Python

1. HOW A COMPUTER WORKS

A computer is an electronic machine that processes information according to instructions.



Example: You enter $20 + 30$ → the computer processes the calculation → you get 50.

2. HARDWARE & SOFTWARE

HARDWARE	SOFTWARE
Physical parts you can interact with.	Programs and instructions.
CPU • RAM • Storage • Screen	Android • Windows • WhatsApp • Python programs

3. PROGRAMMING

Programming is the process of designing and writing instructions a computer can execute to solve problems or perform tasks.

Coding is writing those instructions using a programming language.



4. COMPUTERS FOLLOW INSTRUCTIONS

A computer does not automatically understand what you meant. It follows the instructions it receives.

Think like a programmer:
If the instructions are wrong, the result can be wrong—even if the computer followed the instructions correctly.

5. ALGORITHMS

An algorithm is a clear, step-by-step procedure for solving a problem or completing a task.

Example: Making tea

1	Get a cup.
2	Get water and boil it.
3	Put tea in the cup.
4	Pour the hot water into the cup.
5	Add sugar.
6	Stir.

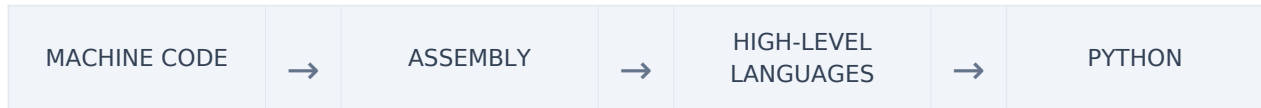


6. WHY PROGRAMMING LANGUAGES?

Computers ultimately execute machine-level instructions, but humans need practical ways to express complex instructions.

Programming languages give humans a more usable way to describe what a computer should do.

7. FROM MACHINE CODE TO PYTHON



Low-level languages: closer to the machine.

High-level languages: provide more human-friendly abstractions.

The important idea is not to memorize old languages. It is to understand that programming languages evolved to help humans work with computers more effectively.

8. WHERE PYTHON CAME FROM

Python was created by Guido van Rossum. Work began around 1989 and an early version was released publicly in 1991. Python grew from Guido's experience with the ABC language.

The name "Python" came from Monty Python's Flying Circus.

9. HOW PYTHON CODE RUNS

Your computer does not directly understand Python source code. A Python implementation/runtime processes it so the computer can execute the program.



You do not need to understand everything happening underneath yet. We will learn what we need as we build.

10. THE BUILDING BLOCKS

Concept	Think of it as...
Input	Information coming into a program
Output	Information a program gives back
Variables	Names attached to values we want to use
Data	Information such as text, numbers, scores
Conditions	Making decisions
Loops	Repeating instructions
Functions	Reusable instructions for a specific job

11. YOUR PROGRAMMER MINDSET

**1. Programming is problem-solving.
Start with the problem, not the syntax.**

**2. Syntax is not the whole game.
Knowing Python words is not the same as knowing how to solve problems.**

**3. Errors are information.
Debugging is a normal part of programming.**

**4. Computers are literal.
Clear instructions and clear logic matter.**

**5. You do not need to know everything first.
Learn the foundation, then build.**

YOUR MINI CHALLENGE

Design the steps for a program that checks whether someone is eligible to vote based on age.

Do not write Python yet. Write only the instructions.

1. _____

2. _____

3. _____

4. _____

5. _____

NEXT SESSION

Next, we turn these ideas into actual Python code.

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